

Tuncay YAYLALI

Data & DataOps Engineer |
Environmental Engineerig, Business
Administration & MIS Professional



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About

Inspired by a deep appreciation for the digital world and its limitless potential for innovation and growth, transitioned to IT sector as a **Data Engineer** in 2022, building on 24 years of experience in Environmental Engineering across both public and private sectors.

Specialized in designing and optimizing **scalable data solutions** for cloud and on-premises environments, with expertise in **data pipelines, infrastructure management**, and end-to-end **system development for real-time and batch processing**.

Certified as an **AWS Solutions Architect Associate** and **SnowPro Core**.

Please note, my CV separates experience in the IT and Environmental sectors for clarity, highlighting my transition and contributions in both fields.

Education

Management Information Systems (B.Sc)

Istanbul University

09.2020 – 01.2025 | Istanbul, Turkey

Occupational Safety (M.Sc)

Marmara University

10.2020 – 07.2021 | Istanbul, Turkey

Enviromental Management (M.Sc)

Eskisehir Technical University

01.2019 – 02.2020 | Eskisehir, Turkey

Business Administration (B.Sc)

Anadolu University

09.2017 – 09.2020 | Eskisehir, Turkey

Environmental Engineering (B.Sc)

Istanbul Technical University

08.1988 – 07.1993 | Istanbul, Turkey

Languages

Turkish: Native

English: B2

German: A2

Professional Experience (IT Sector/2022-Present)

▲ PRACTICUS AI LLC

Data Engineer

02.2025 – 07.2025 | Ankara (Remote), Turkey

- Contributed to the successful deployment and configuration of the **Practicus AI Data Intelligence Platform** across various environments, including on-premise **OpenShift** clusters and major **cloud vendors**.
- Assisted customers in building robust and scalable **data pipelines** using the platform, collaborating to understand their data architecture needs and customize solutions.
- Supported the implementation and management of comprehensive **monitoring, logging, and alerting** solutions using tools like **Prometheus, Grafana, and Fluentd**.
- Contributed to the **troubleshooting** and resolution of complex platform-related issues.
- Assisted in preparing and delivering technical presentations for **pre-sales activities** and developed detailed **technical documentation**, including installation guides and best practices.

▲ DATALONGA INFORMATION TECHNOLOGIES Inc.

Data Engineer

12.2023 – 01.2025 | Istanbul (Hybrid), Turkey

- Managed deployment, configuration, and monitoring of **Rancher RKE2 Kubernetes** clusters for a major telecom company, contributing to a Proof of Concept (PoC) project that achieved an **85% success rate**. In this scope;
 - Implemented the cluster on bare-metal servers using **CEPH** for block and object storage.
 - Integrated and configured 20 fully **open-source tools**.
 - Developed a streaming data pipeline where simulated data from **Redpanda** was processed via **Spark Operator**, stored in **CEPH** Object Storage, cataloged with **Nessie**, and queried through **Trino**, with visualizations created in **Superset**.
 - Monitored the entire system with **Prometheus** and **Grafana** for realtime performance insights.
- Developed a fully open-source **Data Lakehouse** product capable of running in any **Kubernetes** environment, reducing deployment time by **90%** through automated configuration management and enhancing scalability.
- Supported ETL pipelines for one of Turkey's largest electric scooter companies, resolving data inconsistencies using **AWS S3, dbt Cloud, Snowflake**, and **BigQuery**, and enabling access to **17 days of critical missing data**. This improved reporting accuracy by **100%**.
- Flattened complex nested data structures from a **MongoDB** database and successfully transferred them to **BigQuery** using **Apache Airflow**. This process reduced query execution time by **50%** and significantly enhanced the visualization of essential business processes.
- Delivered **6 hours of training** on **Python for Basic Data Analysis** to dozens of students as a **trainer** in the **Data101 Bootcamp**.
- Provided **12 hours of training** on **Snowflake, Databricks, Terraform, and Storage Types** (block storage, object storage, file storage) to hundreds of students as a **trainer** in the **Data Engineering Bootcamp**.
- Acted as a **mentor** to hundreds of students in the **Data101, Data Engineering, and MLOps Bootcamps**, offering technical guidance and career advice throughout the programs.
- Successfully earned the **AWS Solutions Architect Associate Certification** on the first attempt.

▲ VODAFONE TURKEY/via BENTEGO

Data Engineer

01.2023 – 04.2023 | Istanbul (Remote), Turkey

- Utilized Agile methodologies to deliver data engineering solutions, completing **20** tasks across **4** sprints with a **95%** on-time delivery rate.
- Designed and optimized **data flows** and **ELT pipelines** on **Cloudera** to improve overall data processing performance, streamlined model scheduling and monitoring processes to enhance workflow reliability, and collaborated with cross-functional teams to improve pipeline efficiency and scalability.

▲ BENTEGO TECHNOLOGY Inc.

Data Engineer

11.2022 – 04.2023 | Istanbul (Hybrid), Turkey

- As an **outsourced data engineer**, provided customized data engineering solutions to clients, effectively addressing their specific needs and challenges.
- Gained extensive **hands-on experience** and **in-service training** in **big data technologies**, including **data pipelines, storage systems, and analytics platforms**.
- Successfully earned the **SnowPro Core Certification** on the first attempt.

Hard Skills

Programming and Scripting

- Python
- Shell Scripting
- SQL & T-SQL

Operating Systems

- Linux

Database and Data Storage

- Relational Databases: PostgreSQL, MySQL, MS SQL
- NoSQL Databases: Apache Cassandra, MongoDB, Elasticsearch
- Data Lakes and Storage: Delta Lake, Snowflake, CEPH (Block and Object Storage)
- Big Data Databases: HBase, Impala

Data Processing and Big Data Technologies

- Apache Spark, Spark Streaming
- Apache Hadoop (HDFS, YARN, and MapReduce)
- Apache Hive
- Apache Kafka
- Apache Iceberg
- Pandas, NumPy, PyArrow, and PySpark

Data Flow and Workflow Management

- Apache Airflow
- dbt Core

DevOps and CI/CD Tools

- Docker
- Kubernetes
- Terraform
- Ansible
- Jenkins
- GitHub Actions
- Rancher
- Helm
- Kustomize

Cloud Services

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)

Machine Learning and Web API

- Machine Learning
- FastAPI
- MLflow

Environmental Compliance & Assessments

- Environmental & Social Impact Assessment (ESIA), Gap Analysis, and Environmental and Social Action Plans
- Environmental Impact Assessment (EIA/ÇED)

Environmental Modeling & GIS

- Air Quality Dispersion Modeling (ISCST, ISCLT, Aermol)
- Spatial Mapping
- GIS Applications (ArcGIS, ArcMap, Global Mapper, Netcad)

Data Management & Systems

- Relational Database Design
- Data Infrastructure Management
- Quality Management Systems (QMS)

Project Leadership

- Multidisciplinary Team Management
- Regulatory Framework Coordination
- International Liaison & Representation

Key Certification (IT Sector)

Cloud Engineer Bootcamp (Istanbul Data Science Academy)

10.2023 – 07.2024

Kubernetes Bootcamp (Veri Bilimi Okulu)

03.2024 – 06.2024

AWS Certified Solutions Architect - Associate (SAA) (Amazon Web Services)

10.2023 – 10.2026

SnowPro Core Certification (Snowflake)

11.2022

Data Engineer Path (Miuul)

04.2022 – 09.2022

Machine Learning Operations (MLOps) Bootcamp (Veri Bilimi Okulu)

02.2022 – 06.2022

Data Science Specialization Training (Data MasTR) (Advancity and Miuul (in cooperation with Enocta))

10.2021 – 03.2022

Other Certificates

04.2021 – present

A comprehensive overview of my certifications and achievements in the IT sector can be found on my [LinkedIn profile](#) .

Professional Experience (Environmental Sector/1998-2022)

Savra Cons. Mach. Eng. Env. Consultancy Ind. Trade Ltd. Co.

Environmental Engineer (Technical Coordinator)

03.2020 – 10.2022 | Ankara, Turkey

- Managed comprehensive Environmental Impact Assessment (EIA) projects for large-scale metallic mining operations, specifically for clients listed on the BIST500, a major Turkish stock market index.
- Oversaw the full cycle of air quality dispersion modeling and spatial mapping projects, providing technical guidance through in-house training and managing a multidisciplinary team of experts.
- Coordinated and oversaw the preparation of Gap Analysis, ESIA (Environmental and Social Impact Assessment), and Environmental and Social Action Plans for a mining project under one of Turkey's leading industrial groups, ensuring full compliance with IFC Performance Standards and managing the submission process to international lending institutions.

Hidromark Eng. Consultancy Co.

Environmental Engineer (Department Coordinator)

11.2018 – 02.2020 | Ankara, Turkey

- Led EIA processes for large-scale dam and pond construction projects commissioned by the State Hydraulic Works (DSİ), acting as the prime contractor.

MGS Project Consultancy Eng. Ltd. Co.

Environmental Engineer (General Coordinator)

11.2016 – 08.2018 | Ankara, Turkey

- Managed comprehensive EIA processes for diverse sectors, leading a large, multidisciplinary project team through complex environmental regulatory frameworks.
- Executed air quality dispersion modeling and spearheaded technical designs for solid waste sanitary landfills and mine tailings storage facilities.

Duru Env. Tech. and Lab. Serv. Eng. Consultancy Trade Ind. and Trade. Ltd. Co.

Environmental Engineer (Company Manager)

02.2007 – 11.2016 | Ankara, Turkey

- As a Co-founder, established the company as a key player in the environmental engineering sector.
- Spearheaded large-scale projects, including complex industrial EIA processes and solid waste management facilities, while creating jobs and cultivating a team of high-performing multidisciplinary professionals.

Serdar Eng. Consultancy. Cons.. Ind. and Trade. Ltd. Co.

Environmental Engineer (Department Coordinator)

08.2005 – 01.2007 | Ankara, Turkey

- Led successful EIA processes for numerous high-profile projects within the cement industry.
- Recognized as a top-tier expert in air quality dispersion modeling, specializing in the complex conversion and processing of meteorological data for ISCST and ISCLT models.



Member Associations and Organizations

Informatics Association of Turkey

Member

2022 – present | Istanbul, Turkey

Association of ITU Alumni

Member

2020 – present | Istanbul, Turkey

Chamber of Environmental Engineers

Member

1998 – present | Ankara, Turkey



Interests

Content Creation on Social Media

- Regularly create and design engaging posts for platforms such as **LinkedIn** and **X** (formerly Twitter). Focus on topics like technology trends, data engineering, and professional development.

Writing on Medium (IT and Tech)

- Publish insightful articles on **Medium** about IT-related topics, including data engineering, cloud solutions, and modern software practices. Passionate about breaking down complex technical concepts into accessible and engaging narratives.

Exploring Emerging Technologies

- Actively explore and share insights on the latest trends in the IT industry, such as cloud computing, DevOps, and data engineering. Leverage resources like **Coursera** , **LinkedIn Learning** , **Udemy** , **DataCamp** , **aciLearning** , **Pluralsight** , **Manning** , **O'Reilly** and **Packt** to stay updated and continuously enhance knowledge.



Hobbies

Cooking

- Baking pastries and dairy desserts

Sport

- Going to the gym regularly every week, focusing on weightlifting, and occasionally playing basketball.

Music

- Listening to Loreena McKennitt's Celtic melodies and Ramin Djawadi's cinematic compositions.

Outdoor Activities and Travelling

- Hiking in nature, especially on sunny days, and engaging in amateur climbing on mountainous terrains whenever possible.
- Traveling, particularly long road trips by car, discovering new places, and embracing unique experiences in different locations.

Atak Tours & Travel Congress, Seminars, Meetings & Organizations

Project Manager

09.2004 – 07.2005 | Ankara, Turkey

- Managed the robust data infrastructure for large-scale national events with tens of thousands of participants.
- Designed and implemented comprehensive Quality Management Systems (QMS), streamlining operational efficiency and documentation standards.

Ministry of Environment and Forestry

Environmental Engineer

12.1998 – 09.2004 | Ankara, Turkey

- Spearheaded the national implementation of international nature protection conventions, including the Bern Convention and CITES, while serving as the official representative for Turkey in international environmental forums.
- Architected and developed a comprehensive relational database for Turkey's endemic plant species, ensuring data integrity and accessibility for conservation research.
- Designed and developed of the official website for the General Directorate of Nature Conservation and National Parks.